

MATH3026/4022 Lecture 5

Autoregressive Moving Average (ARMA)

Processes and the Autocovariance Generating Function for Time Series

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12th February 2026

- ▶ Definition of the autoregressive moving average (ARMA) process.
- ▶ The $ARMA(1, 1)$ process:
 - ▶ Moving average form
 - ▶ Autocorrelation function
- ▶ General properties of ARMA processes
- ▶ Autocovariance generating function (AGF)

Autoregressive Moving Average (ARMA) Processes

Having explored both autoregressive ($AR(p)$) processes and moving average ($MA(q)$) processes, we now consider a useful class of time series model known as an **autoregressive moving average** process.

Essentially, the **autoregressive moving average process** is a process that includes both autoregressive and moving average terms.

This process can be a convenient model for data where we suspect that there is dependence on past values of the series **and** on past error terms.

Autoregressive Moving Average (ARMA) Processes

Suppose $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 . Then $\{X_t : t \in \mathbb{Z}\}$ is an **autoregressive moving average** (ARMA) process of order (p, q) if

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}.$$

Usually, we refer to this process as an $ARMA(p, q)$ process.

Here

- ▶ p = order of the autoregressive (AR) part of the process;
- ▶ q = order of the moving average (MA) part of the process.

Autoregressive Moving Average (ARMA) Processes

Using the backward shift operator, B , we can write the $ARMA(p, q)$ process as

$$\phi(B)X_t = \theta(B)Z_t$$

where

$$\begin{aligned}\phi(B) &= 1 - \phi_1 B - \dots - \phi_p B^p; \\ \theta(B) &= 1 + \theta_1 B + \dots + \theta_q B^q.\end{aligned}$$

We note that the values of $\{\phi_j\}$ that ensure that the $ARMA(p, q)$ process is (weakly) stationary are those such that the roots of

$$\phi(z) = 0$$

lie outside the unit circle.

Autoregressive Moving Average (ARMA) Processes

Moreover, the values of $\{\theta_j\}$ that ensure that the $ARMA(p, q)$ process is invertible are those such that the roots of

$$\theta(z) = 0$$

lie outside the unit circle.

We note that both the $MA(q)$ and $AR(p)$ processes are special cases of an $ARMA(p, q)$ process since

$$\begin{aligned}\theta_j &= 0 \text{ for all } j \implies ARMA(p, q) \equiv AR(p) \\ \phi_j &= 0 \text{ for all } j \implies ARMA(p, q) \equiv MA(q).\end{aligned}$$

First, we will explore the $ARMA(1, 1)$ process before examining properties of $ARMA$ processes of higher order.

The $ARMA(1,1)$ Process

We consider the $ARMA(1,1)$ process written

$$\begin{aligned} X_t &= \phi_1 X_{t-1} + Z_t + \theta_1 Z_{t-1} \\ \implies X_t - \phi_1 X_{t-1} &= Z_t + \theta_1 Z_{t-1} \end{aligned} \tag{5.1}$$

with $|\phi_1| < 1$ and $|\theta_1| < 1$ to ensure (weak) stationarity and invertibility, respectively.

Equation (5.1) may be rewritten

$$(1 - \phi_1 B)X_t = (1 + \theta_1 B)Z_t.$$

It follows that

$$X_t = (1 - \phi_1 B)^{-1}(1 + \theta_1 B)Z_t.$$

The $ARMA(1,1)$ process - MA form

Then

$$\begin{aligned}X_t &= (1 - \phi_1 B)^{-1} (1 + \theta_1 B) Z_t \\&= (1 - \phi_1 B)^{-1} (1 - \phi_1 B + \phi_1 B + \theta_1 B) Z_t \quad (\text{just adding } 0) \\&= (1 - \phi_1 B)^{-1} [(1 - \phi_1 B) + (\phi_1 + \theta_1)B] Z_t \\&= \left[1 + (\phi_1 + \theta_1) \frac{B}{1 - \phi_1 B} \right] Z_t \\&= Z_t + (\phi_1 + \theta_1) \sum_{j=0}^{\infty} \phi_1^j B^{j+1} Z_t \\&= Z_t + (\phi_1 + \theta_1) \sum_{j=1}^{\infty} \phi_1^{j-1} Z_{t-j}.\end{aligned}$$

Since the Taylor series expansion of $\frac{1}{1-\phi_1 B}$ is $\sum_{j=0}^{\infty} \phi_1^j B^j$.

The $ARMA(1,1)$ process - MA form

Hence we see that the $ARMA(1,1)$ process may be written in $MA(\infty)$ form as

$$X_t = \sum_{j=0}^{\infty} \alpha_j Z_{t-j}$$

with

$$\alpha_0 = 1;$$

$$\alpha_j = (\phi_1 + \theta_1)\phi_1^{j-1} \quad j = 1, 2, \dots$$

$ARMA(1,1)$ process: Autocorrelation function

Now we consider the autocorrelation function for the $ARMA(1,1)$ process.

As with the $AR(p)$ process (Lecture 4), let us assume that the $ARMA(1,1)$ process is (weakly) stationary.

We begin with

$$X_t - \phi_1 X_{t-1} = Z_t + \theta_1 Z_{t-1}. \quad (5.2)$$

Next, we multiply both sides of (5.2) by Z_t and take expectations

$$\mathbb{E}(X_t Z_t) - \phi_1 \mathbb{E}(X_{t-1} Z_t) = \mathbb{E}(Z_t^2) + \theta_1 \mathbb{E}(Z_{t-1} Z_t).$$

$ARMA(1,1)$ process: Autocorrelation function

We note that

$$\mathbb{E}(X_{t-1}Z_t) = 0$$

because of our causality assumption (X_t is independent from future white noise terms). Also

$$\mathbb{E}(Z_{t-1}Z_t) = 0 \quad (\text{because } Z_{t-1} \text{ and } Z_t \text{ are independent}).$$

Hence

$$\begin{aligned}\mathbb{E}(X_t Z_t) - \phi_1 \mathbb{E}(X_{t-1} Z_t) &= \mathbb{E}(Z_t^2) + \theta_1 \mathbb{E}(Z_{t-1} Z_t) \\ \implies \mathbb{E}(X_t Z_t) &= \mathbb{E}(Z_t^2) = \sigma_z^2.\end{aligned}$$

ARMA(1,1) process: Autocorrelation function

Then, multiplying both sides of (5.2) by X_t and taking expectations, we obtain

$$\begin{aligned}\mathbb{E}(X_t^2) - \phi_1 \mathbb{E}(X_t X_{t-1}) &= \mathbb{E}(X_t Z_t) + \theta_1 \mathbb{E}(X_t Z_{t-1}) \\ \gamma(0) - \phi_1 \gamma(1) &= \sigma_z^2 + \theta_1 \mathbb{E}(X_t Z_{t-1})\end{aligned}\tag{5.3}$$

Next, to find $\mathbb{E}(X_t Z_{t-1})$, we multiply (5.2) by Z_{t-1} and take expectations of both sides, whence

$$\begin{aligned}\mathbb{E}(X_t Z_{t-1}) - \phi_1 \mathbb{E}(X_{t-1} Z_{t-1}) &= \mathbb{E}(Z_t Z_{t-1}) + \theta_1 \mathbb{E}(Z_{t-1}^2) \\ \implies \mathbb{E}(X_t Z_{t-1}) - \phi_1 \sigma_z^2 &= 0 + \theta_1 \sigma_z^2 \\ \implies \mathbb{E}(X_t Z_{t-1}) &= \sigma_z^2 (\phi_1 + \theta_1).\end{aligned}$$

$ARMA(1,1)$ process: Autocorrelation function

Therefore, substituting into (5.3), we obtain

$$\begin{aligned}\gamma(0) - \phi_1\gamma(1) &= \sigma_z^2 + \sigma_z^2\theta_1(\phi_1 + \theta_1) \\ \implies \gamma(0) &= \phi_1\gamma(1) + \sigma_z^2(1 + \theta_1(\phi_1 + \theta_1)).\end{aligned}\quad (1)$$

Recalling (5.2) (re-written below)

$$X_t - \phi_1 X_{t-1} = Z_t + \theta_1 Z_{t-1}$$

We multiply both sides of the above equation by X_{t-1} and take expectations.

$$\begin{aligned}\mathbb{E}(X_t X_{t-1}) - \phi_1 \mathbb{E}(X_{t-1}^2) &= \mathbb{E}(X_{t-1} Z_t) + \theta_1 \mathbb{E}(X_{t-1} Z_{t-1}) \\ \implies \gamma(1) - \phi_1 \gamma(0) &= 0 + \theta_1 \sigma_z^2 \\ \implies \gamma(1) &= \phi_1 \gamma(0) + \theta_1 \sigma_z^2.\end{aligned}\quad (2)$$

ARMA(1, 1) process: Autocorrelation function

Equations (1) and (2) are a pair of simultaneous equations for $\gamma(0)$ and $\gamma(1)$, which we'll now solve.

$$\gamma(0) = \phi_1 \gamma(1) + \sigma_z^2 (1 + \theta_1(\phi_1 + \theta_1)) \quad (1)$$

$$\gamma(1) = \phi_1 \gamma(0) + \theta_1 \sigma_z^2 \quad (2)$$

$ARMA(1, 1)$ process: Autocorrelation function

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$ARMA(1,1)$ process: Autocorrelation function

We see that, for the $ARMA(1,1)$ process,

$$\gamma(h) = \begin{cases} \frac{\sigma_z^2(1+2\theta_1\phi_1+\theta_1^2)}{1-\phi_1^2} & h = 0; \\ \frac{\sigma_z^2(1+\theta_1\phi_1)(\phi_1+\theta_1)}{1-\phi_1^2} & |h| = 1; \\ \phi_1^{|h|-1}\gamma(1) & |h| > 1. \end{cases}$$

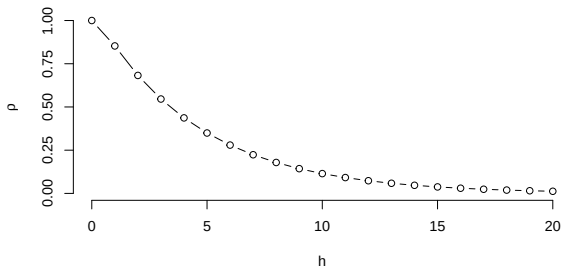
Thus, the ACF ($\rho(h) = \gamma(h)/\gamma(0)$) is given by

$$\rho(h) = \begin{cases} \frac{(1+\theta_1\phi_1)(\phi_1+\theta_1)}{1+2\theta_1\phi_1+\theta_1^2} & |h| = 1; \\ \phi_1^{|h|-1}\rho(1) & |h| > 1. \end{cases}$$

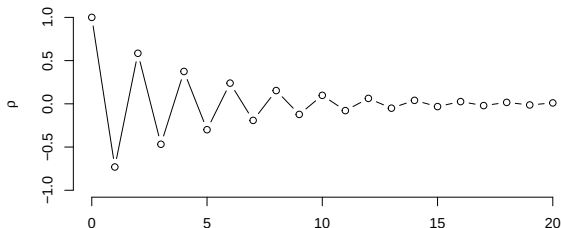
We note that $\rho(h) \rightarrow 0$ as $h \rightarrow \infty$ since $|\phi_1| < 1$.

ARMA(1,1) process: Autocorrelation function

ACF for ARMA(1,1) process, $\phi_1 = 0.8$, $\theta_1 = 0.2$



ACF for ARMA(1,1) process, $\phi_1 = -0.8$, $\theta_1 = 0.15$



$ARMA(p, q)$ process: Autocorrelation function

More generally, for the $ARMA(p, q)$ process:

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

Assuming (weak) stationarity, multiplying the above by X_{t-h} ($h > 0$) and taking expectations, we obtain

$$\begin{aligned}\mathbb{E}(X_t X_{t-h}) &= \phi_1 \mathbb{E}(X_{t-1}, X_{t-h}) + \dots + \phi_p \mathbb{E}(X_{t-p}, X_{t-h}) \\ &\quad + \mathbb{E}(X_{t-h} Z_t) + \theta_1 \mathbb{E}(X_{t-h} Z_{t-1}) + \dots + \theta_q \mathbb{E}(X_{t-h} Z_{t-q}) \\ \implies \gamma(-h) &= \phi_1 \gamma(-h+1) + \dots + \phi_p \gamma(-h+p) \\ &\quad + \mathbb{E}(X_{t-h} Z_t) + \theta_1 \mathbb{E}(X_{t-h} Z_{t-1}) + \dots + \theta_q \mathbb{E}(X_{t-h} Z_{t-q}).\end{aligned}$$

$ARMA(p, q)$ process: Autocorrelation function

Because of the stationarity assumption

$$\begin{aligned}\gamma(h) &= \phi_1\gamma(h-1) + \dots + \phi_p\gamma(h-p) \\ &\quad + \mathbb{E}(X_{t-h}Z_t) + \theta_1\mathbb{E}(X_{t-h}Z_{t-1}) + \dots + \theta_q\mathbb{E}(X_{t-h}Z_{t-q})\end{aligned}$$

where

$$\mathbb{E}(X_{t-h}Z_{t-k}) \begin{cases} = 0 & \text{if } h > k; \\ \neq 0 & \text{if } h \leq k. \end{cases}$$

since past values of the process do not influence future white noise terms (causality assumption).

For the same reason, if $h > q$ then $\mathbb{E}(X_{t-h}Z_{t-k}) = 0$ for all k since $0 \leq k \leq q < h$ (i.e. h is always greater than k).

ARMA(p, q) process: Autocorrelation function

Hence, if $h \geq q + 1$ then

$$\gamma(h) = \phi_1 \gamma(h-1) + \dots + \phi_p \gamma(h-p)$$

and, dividing the above by $\gamma(0)$,

$$\rho(h) = \phi_1 \rho(h-1) + \dots + \phi_p \rho(h-p).$$

Hence, we have a method to find $\rho(h)$ for $h \geq q + 1$, where $\rho(1), \dots, \rho(q)$ are known.

However, in practice, determining $\rho(1), \dots, \rho(q)$ may involve solving difficult sets of equations, especially when q is large.

As a result, we would usually prefer to fit ARMA models of a low order (i.e. where p and q are small).

$ARMA(p, q)$ process - different forms

Recall that the $ARMA(p, q)$ process can be written

$$\phi(B)X_t = \theta(B)Z_t$$

where

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p;$$

$$\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q.$$

It follows that the $ARMA(p, q)$ process can be written as a general linear process (i.e. in $MA(\infty)$ form) because

$$\begin{aligned} X_t &= [\phi(B)]^{-1} \theta(B) Z_t \\ &= \alpha(B) Z_t \\ &= \sum_{j=0}^{\infty} \alpha_j B^j Z_t = \sum_{j=0}^{\infty} \alpha_j Z_{t-j} \end{aligned}$$

for suitable weights $\{\alpha_j\}$.

$ARMA(p, q)$ process - different forms

Moreover, we can also express an $ARMA(p, q)$ process in pure AR form since

$$\begin{aligned}Z_t &= [\theta(B)]^{-1} \phi(B) X_t \\ &= \pi(B) X_t\end{aligned}$$

where

$$\pi(B) = \sum_{j=0}^{\infty} \pi_j B^j$$

for suitable terms $\{\pi_j\}$.

We see that

$$X_t = \sum_{j=1}^{\infty} \pi_j X_{t-j} + Z_t \quad (\text{setting } \pi_0 = 1)$$

which is an AR process of infinite order.

Example 5.1

The process $\{X_t\}$ is an $ARMA(1, 1)$ process where

$$X_t = \frac{1}{2}X_{t-1} + Z_t - \frac{3}{10}Z_{t-1}.$$

- (a) Verify that the process is both (weakly) stationary and invertible.
- (b) Write the process as a moving average process.
- (c) Write the process as an autoregressive process.

Example 5.1

Example 5.1

Autocovariance Generating Function

So far, we have introduced some common time series models and explored various model properties.

This has included calculation of the autocorrelation function (ACF) for different model types.

We saw that calculating the ACF often involved solving sets of equations and/or a substantial number of calculations.

Naturally, it would be helpful if we could use a more straightforward method to calculate the ACF for a process.

To do this, we introduce the **autocovariance generating function** (AGF) which can often yield an easier approach to calculating autocovariance (and hence autocorrelation).

Definition 5.1: Autocovariance Generating Function

Let $\{X_t : t \in \mathbb{Z}\}$ be a (weakly) stationary process. The autocovariance generating function (AGF) of the process is defined as the scalar-valued function $\Gamma(z)$ where

$$\Gamma(z) = \sum_{h=-\infty}^{\infty} \gamma(h)z^h.$$

The above definition implies that if we can compute $\Gamma(z)$ as a polynomial with terms z , z^2 etc. then the autocovariance at lag h is given by the coefficient of z^h .

General Linear Process: AGF

Hence, if we can find a way by which $\Gamma(z)$ can be determined easily, then we can obtain the autocovariance (and thus the autocorrelation) at lag h with ease.

We have seen that the common time series models encountered (e.g. $MA(q)$, $AR(p)$ and $ARMA(p, q)$) can be written as general linear processes (i.e. in $MA(\infty)$ form).

That is, we can write these common time series in the form

$$X_t = \sum_{j=0}^{\infty} \alpha_j Z_{t-j} = \alpha(B) Z_t$$

where

$$\alpha(B) = \alpha_0 + \alpha_1 B + \alpha_2 B^2 + \dots$$

and $\{Z_t\}$ is a white noise process with variance σ_z^2 .

General Linear Process: AGF

In Lecture 3 we have seen that

$$\mathbb{E}(X_t X_{t+h}) = \gamma(h) = \sigma_z^2 \sum_{j=0}^{\infty} \alpha_j \alpha_{j+h}.$$

It follows that the AGF of the $MA(\infty)$ process is given by

$$\begin{aligned} \Gamma(z) &= \sigma_z^2 \sum_{h=-\infty}^{\infty} \left(\sum_{j=0}^{\infty} \alpha_j \alpha_{j+h} \right) z^h \\ &= \sigma_z^2 \sum_{j=0}^{\infty} \sum_{h=-j}^{\infty} \alpha_j \alpha_{j+h} z^h \end{aligned} \tag{5.4}$$

Because $\alpha_{j+h} = 0$ where $j+h < 0$ (i.e. where $h < -j$).

If we set $j + h = k$, then Equation (5.4) becomes

$$\begin{aligned}\Gamma(z) &= \sigma_z^2 \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \alpha_j \alpha_k z^{k-j} \\ &= \sigma_z^2 \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \alpha_j z^{-j} \alpha_k z^k \\ &= \sigma_z^2 \left(\sum_{k=0}^{\infty} \alpha_k z^k \right) \left(\sum_{j=0}^{\infty} \alpha_j z^{-j} \right) \\ &= \sigma_z^2 \alpha(z) \alpha(z^{-1})\end{aligned}$$

recalling that $\alpha(z) = \alpha_0 + \alpha_1 z + \alpha_2 z^2 + \dots$

Autocovariance Generating Function

We see that we can

1. Write a time series, X_t , in $MA(\infty)$ form

$$X_t = \sum_{j=0}^{\infty} \alpha_j Z_{t-j} = \alpha(B)Z_t$$

2. Determine the polynomial $\alpha(z)$.
3. Determine $\Gamma(z) = \sigma_z^2 \alpha(z)\alpha(z^{-1})$.
4. Deduce that the autocovariance at lag h (i.e. $\gamma(h)$) is the coefficient of z^h in $\Gamma(z)$.

Example 5.2

Determine the autocovariance generating function for the $MA(1)$ process.

$$X_t = \theta Z_{t-1} + Z_t.$$

AGF for the $ARMA(p, q)$ process

Recall that the $ARMA(p, q)$ process may be written

$$\phi(B)X_t = \theta(B)Z_t \quad (5.5)$$

where

$$\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

$$\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$$

Rearranging Equation (5.5), we see that

$$\begin{aligned} X_t &= \phi(B)^{-1} \theta(B) Z_t \\ &= \alpha(B) Z_t \quad (\text{where } \alpha(B) = \phi(B)^{-1} \theta(B)). \end{aligned}$$

AGF for the $ARMA(p, q)$ process

We deduce that the autocovariance generating function for the $ARMA(p, q)$ process is

$$\begin{aligned}\Gamma(z) &= \sigma_z^2 \alpha(z) \alpha(z^{-1}) \\ &= \sigma_z^2 \frac{\theta(z) \theta(z^{-1})}{\phi(z) \phi(z^{-1})}.\end{aligned}$$

Assuming that the ratio of the polynomials $\theta(z)$ and $\phi(z)$ can be calculated easily, we will be able to deduce that the coefficient of z^h is the autocovariance at lag h , $\gamma(h)$.

Lecture 5: Learning Outcomes

- ▶ Know the definition of the autoregressive moving average (ARMA) process.
- ▶ Understand how to derive and interpret the autocorrelation function for the $ARMA(1, 1)$ process.
- ▶ Understand the form of the autocorrelation function for the more general $ARMA(p, q)$ process.
- ▶ Know the definition of the autocovariance generating function (AGF) and understand how this function may be used to determine autocovariance at lag h for a time series.